

Chapter Five- Analysis of Variance, Department of Accounting and Finance Set by: Melaku Y.

F distribution

d.f.D.: degrees of freedom, denominator	$\alpha = 0.05$																	
	d.f.N.: degrees of freedom, numerator																	
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22

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Chapter Five

Analysis of Variance

When an F test is used to test a hypothesis concerning the means of three or more populations, the technique is called **analysis of variance** (commonly abbreviated as ANOVA).

Assumptions for the F Test

1. The populations from which the samples were obtained must be normally or approximately normally distributed.
2. The samples must be independent of one another.
3. The variances of the populations must be equal.

Even though you are comparing three or more means in this use of the F test, *variances* are used in the test instead of means.

With the F test, two different estimates of the population variance are made. The first estimate is called the **between-group variance**, and it involves finding the variance of the means. The second estimate, the **within-group variance**, is made by computing the variance using all the data and is not affected by differences in the means. If there is no difference in the means, the between-group variance estimate will be approximately equal to the within-group variance estimate, and the F test value will be approximately equal to 1. The null hypothesis will not be rejected.

However, when the means differ significantly, the between-group variance will be much larger than the within-group variance; the F test value will be significantly greater than 1; and the null hypothesis will be rejected. Since variances are compared, this procedure is called *analysis of variance* (ANOVA).

For a test of the difference among three or more means, the following hypotheses should be used:

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

H_a : At least one mean is different from the others.

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As stated previously, a significant test value means that there is a high probability that this difference in means is not due to chance, but it does not indicate where the difference lies.

The degrees of freedom for this F test are $d.f.N(V1) = k - 1$, where k is the number of Groups, and $d.f.D(V2) = N - k$, where N is the sum of the sample sizes of the groups $N = n1 + n2 + \dots + nk$. The sample sizes need not be equal. The F test to compare means is always right-tailed.

Steps followed in ANOVA

Step 1: State the hypotheses and identify the claim.

Step 2: Find the critical value.

$$V1 = K - 1$$

$$V2 = N - K$$

Step 3: Compute the test value, using the procedure outlined here.

- a) Find the mean and variance of each sample.
- b) Find the grand mean. The *grand mean*, denoted by $\bar{X}_{GM}$ is the mean of all values in the samples.

$$\bar{X}_{GM} = \frac{\sum X}{N}$$

- c) Find the between-group variance, denoted by S_B^2

$$S_B^2 = \frac{\sum ni(\bar{X}_i - \bar{X}_{GM})^2}{K - 1}$$

Note: This formula finds the variance among the means by using the sample sizes as weights and considers the differences in the means.

- d) Find the within-group variance, denoted by S_w^2

$$S_w^2 = \frac{\sum (ni - 1)S_i^2}{\sum (ni - 1)}$$

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Note: This formula finds an overall variance by calculating a weighted average of the individual variances. It does not involve using differences of the means.

e) Find the F test value.

$$F = \frac{S_B^2}{S_w^2}$$

Step 4 Make the decision. By comparing F tabulated and F calculated. If F calculated is greater than F tabulated we will reject the null hypothesis

Illustration: A researcher wishes to try three different techniques to lower the blood pressure of individuals diagnosed with high blood pressure. The subjects are randomly assigned to three groups; the first group takes medication, the second group exercises, and the third group follows a special diet. After four weeks, the reduction in each person's blood pressure is recorded. At a 0.05 level of significance, test the claim that there is no difference among the means. The data are shown below.

Medication	Exercise	Diet
10	6	5
12	8	9
9	3	12
15	0	8
13	2	4
$\bar{X}_1 = 11.8$	$\bar{X}_2 = 3.8$	$\bar{X}_3 = 7.6$
$S_1^2 = 5.7$	$S_2^2 = 10.2$	$S_3^2 = 10.3$

Solution

Step 1: State the hypotheses and identify the claim.

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1 : At least one mean is different from the others.

Step 2: Find the critical value. Since $k = 3$ and $N = 15$,

$$\text{d.f.N.} = k - 1 = 3 - 1 = 2$$

$$\text{d.f.D.} = N - k = 15 - 3 = 12$$

The critical value is 3.89, obtained from table with $\alpha = 0.05$.

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Step 3 Compute the test value.

- a. Find the mean and variance of each sample (these values are shown below the data).
- b. Find the grand mean.

$$= \frac{10 + 12 + 9 + \dots + 4}{15} = \frac{116}{15} = 7.73$$

- c. Find the between-group variance, denoted by S^2_B .

$$\begin{aligned} s_B^2 &= \frac{\sum n_i(\bar{X}_i - \bar{X}_{GM})^2}{k - 1} \\ &= \frac{5(11.8 - 7.73)^2 + 5(3.8 - 7.73)^2 + 5(7.6 - 7.73)^2}{3 - 1} \\ &= \frac{160.13}{2} = 80.07 \end{aligned}$$

- d. Find the within-group variance, denoted by S^2_W .

$$\begin{aligned} s_W^2 &= \frac{\sum (n_i - 1)s_i^2}{\sum (n_i - 1)} \\ &= \frac{(5 - 1)(5.7) + (5 - 1)(10.2) + (5 - 1)(10.3)}{(5 - 1) + (5 - 1) + (5 - 1)} \\ &= \frac{104.80}{12} = 8.73 \end{aligned}$$

- e. Find the F test value.

$$F = \frac{s_B^2}{s_W^2} = \frac{80.07}{8.73} = 9.17$$

Step 4: Make the decision. The decision is to reject the null hypothesis, since $9.17 > 3.89$.

Step 5: Summarize the results. There is enough evidence to reject the claim and conclude that at least one mean is different from the others.